

FitBuddy Project Documentation Phase

Comprehensive Record of Requirements, Architecture, Implementation, User Guidelines, and Maintenance

01. Project Documentation Overview

The project documentation phase is a foundational stage in the software development life cycle of the **FitBuddy** fitness and wellness application [cite: 1, 2]. Its main objective is to establish an exhaustive technical record covering initial planning, core requirements, system architecture, development implementation, QA testing activities, and final outcomes to guide developers, users, evaluators, and future maintainers [cite: 1, 3].

02. Objectives of Project Documentation

Project Information

Provide a clear explanation of the FitBuddy application's purpose and scope [cite: 5].

Requirement Recording

Formally document all functional and non-functional application specifications [cite: 6].

Design Description

Detail the interface layouts, user flow diagrams, and UI navigation components [cite: 7].

Development Details

Log technologies, code structure, tools, and integration methods utilized [cite: 8].

Testing Records

Document test execution scenarios, defect logs, and retesting verifications [cite: 9].

User & Future Reference

Deliver operating manuals for end users and maintainability guides for updates [cite: 10, 11].

03. Project Overview Documentation

Provides core contextual details on FitBuddy, detailing its scope, targeted audience, operational objectives, and key modules such as workout management, fitness goal configuration, activity tracking, and progress charts [cite: 12, 13].

04. Requirement Documentation Breakdown

Requirement Domain	Documented Scope & Deliverables
Functional Requirements	Specifies primary operations, including user onboarding, workout creation, exercise logging, goal updating, and progress visualization [cite: 15, 16].
Non-Functional Requirements	Defines system usability standards, loading speed thresholds, security protocols, reliability, and cross-browser compatibility expectations [cite: 17].
Hardware & Software Specifications	Outlines development hardware alongside programming languages, code editors (VS Code), browser engines, and build environments [cite: 18, 19].

05–09. System Architecture & Lifecycle Deliverables

DOCUMENTATION PILLAR	KEY DELIVERABLES & INCLUDED SPECIFICATIONS
05. Design Documentation	Covers UI layouts, navigation pathways, dashboard architecture, tracking wireframes, responsive viewport breakpoints, and visual user flowcharts [cite: 21, 22, 23].
06. Development Documentation	Logs source code organization, file structures, programming tech stacks, component integration pathways, and version control procedures [cite: 25, 26, 27].
07. Testing Documentation	Catalogues test execution methodologies, test cases, expected vs. actual outcomes, bug reports, debugging activities, and retesting logs [cite: 29, 30, 31].
08. User Manual	Delivers accessible, step-by-step end-user guidelines for navigating the dashboard, customizing workouts, setting goals, and tracking progress [cite: 33, 34, 35].
09. Maintenance Documentation	Provides technical reference for codebase refactoring, known issues, architectural extendibility, and roadmap items for future updates [cite: 36, 37].

10. Expected Outcome

The project documentation phase delivers a complete, well-structured, and permanent reference repository encompassing all lifecycle phases of the FitBuddy application [cite: 38, 39]. It ensures complete transparency for evaluators, seamless onboarding for new developers, simple guidance for end-users, and a stable foundation for long-term maintenance and scalable future feature expansions [cite: 3, 38, 40].